AutoSettle AI Hackathon Project Guide

https://gemini.google.com/app/f749d1b2b19d61eb

Response: Project Overview: AutoSettle AI

AutoSettle AI is a brilliant, high-impact hackathon project. It tackles a massive pain point in the insurance industry (long approval delays and manual inspection bottlenecks) while showcasing advanced technical skills like **Computer Vision (CV)**, **automated cost estimation**, and **instant report generation**.

Given your 7–8 hour timeline, you cannot build a production-grade CV model from scratch. Instead, focus on a **smart MVP (Minimum Viable Product)** using pre-trained models, APIs, or mock inference engines wrapped in a polished, fast UI.

Core Features to Build (Prioritized for 7-8 Hours)

Divide your team and tackle these core modules during the hackathon:

1. **Intelligent Image Upload & Damage Detection Dashboard**
    - **What it does:** A clean, responsive web interface where the user/driver uploads photos of the damaged vehicle (e.g., front bumper, side door, headlight).
    - **Hackathon Shortcut:** Use pre-trained object detection/segmentation libraries or APIs (like YOLOv8/v11, Roboflow, or Hugging Face vision models) to detect dents, scratches, and broken parts, or build a convincing mock inference pipeline that classifies damage severity (Minor, Moderate, Severe).
2. **Automated Severity & Repair Cost Estimator**
    - **What it does:** Maps the detected damage to specific vehicle parts (e.g., bumper, fender, windshield) and calculates an instant estimated repair cost based on predefined part pricing matrices and labor hours.
    - **Hackathon Shortcut:** Create a relational JSON lookup table mapping vehicle parts + damage severity to estimated costs in INR/USD. Display a real-time cost breakdown (Parts + Labor + Tax).
3. **Instant Inspection & Claim Report Generator**
    - **What it does:** Generates a professional, auto-filled PDF/digital claim assessment report containing the uploaded images, AI damage scores, estimated cost breakdown, and a unique **Claim ID**.
    - **Hackathon Shortcut:** Use frontend libraries (like jspdf or HTML-to-print CSS) to let judges download a polished PDF report instantly with one click.
4. **Insurance Desk / Adjuster Review Portal**
    - **What it does:** A separate dashboard view for insurance agents to review incoming claims, view the AI-generated damage score, approve or flag the estimate, and close the claim instantly.

Suggested Tech Stack

- **Frontend:** HTML5, CSS3, JavaScript (Bootstrap 5 or Tailwind CSS) or React for a slick, modern UI.
- **Backend:** Python (Flask or FastAPI) to handle image uploads, process mock/real CV inferences, and generate reports.
- **AI / CV Layer:** OpenCV, Ultralytics YOLO (if running a quick local script), or Roboflow hosted inference API.
- **Report Generation:** Python reportlab or JS jsPDF for instant PDF downloads.

User prompt: it is not mandatory to make full project in 7 to 8 hours we can make main structure this time we have to show that we add important addons and features during hackathon time.

Response: That is a very smart hackathon strategy! In a short 7–8 hour window, judges don't expect a fully deployed, enterprise-grade production software. They want to see **vision, architectural clarity, smart use of AI, and high-impact "killer features" (addons)** that prove you understand the real-world business problem.

Focusing on the **core skeletal structure (skeleton MVP)** plus 2 or 3 standout, high-impact addons will make your project stand out immediately.

Key Addons & Features to Showcase (Judges' Magnets)

Integrating these specialized features into your core structure demonstrates deep thinking and technical ambition:

1. **Fraud Detection & Duplicate Image Guard (The "Anti-Scam" Addon)**
    - **What to show:** People sometimes try to reuse old accident photos or submit edited images for new claims.
    - **How to demo it:** Add a quick perceptual image hashing or metadata check (or a clean UI alert badge) that flags if an uploaded photo has duplicate metadata, altered pixels, or was taken months ago. Judges love this because it solves a multi-billion-dollar insurance fraud problem instantly.
2. **Interactive 3D/2D Car Damage Hotspot Mapper**
    - **What to show:** Instead of just a generic photo upload box, show an interactive wireframe or top-down diagram of a car where users can click or highlight specific damaged zones (e.g., Front Bumper, Left Fender, Rear Door).
    - **How to demo it:** A simple SVG car graphic where clicking a section selects it and links it to the uploaded photo. It looks extremely high-tech and polished for a hackathon.
3. **WhatsApp / SMS Instant Notification Simulator**
    - **What to show:** Real-world insurance apps keep customers updated instantly.
    - **How to demo it:** Build a simulated notification panel or mock API response showing an automated message sent to the driver's phone: *"Claim #AS-9402 Auto-Approved! ₹14,500 transferred to linked garage account."*
4. **Multi-Role Switcher (Customer vs. Insurance Adjuster View)**
    - **What to show:** A toggle switch in your navbar to switch between **"User Mode"** (uploading photos & getting instant quotes) and **"Adjuster Desk"** (reviewing AI scores, modifying payouts, and approving files). This proves you built a complete two-sided workflow.

Suggested Rapid Architecture for 7–8 Hours

- **Core Flow:**
    1. Upload photo →

2. Mock/Real CV bounding-box overlay showing detected dents/scratches ⟶
3. Instant cost calculation table ⟶
4. One-click PDF Claim Report generation.

- **Highlight Addons on Screen:** Add dedicated badge tags or sections labeled "⚡ *Fraud Detection: Verified*", "⚡ *Geo-Tag Check: Passed*", and "⚡ *Instant WhatsApp Alert Triggered*" to catch the judges' eyes immediately during the presentation.

User prompt: can you give me a figma layout to understand

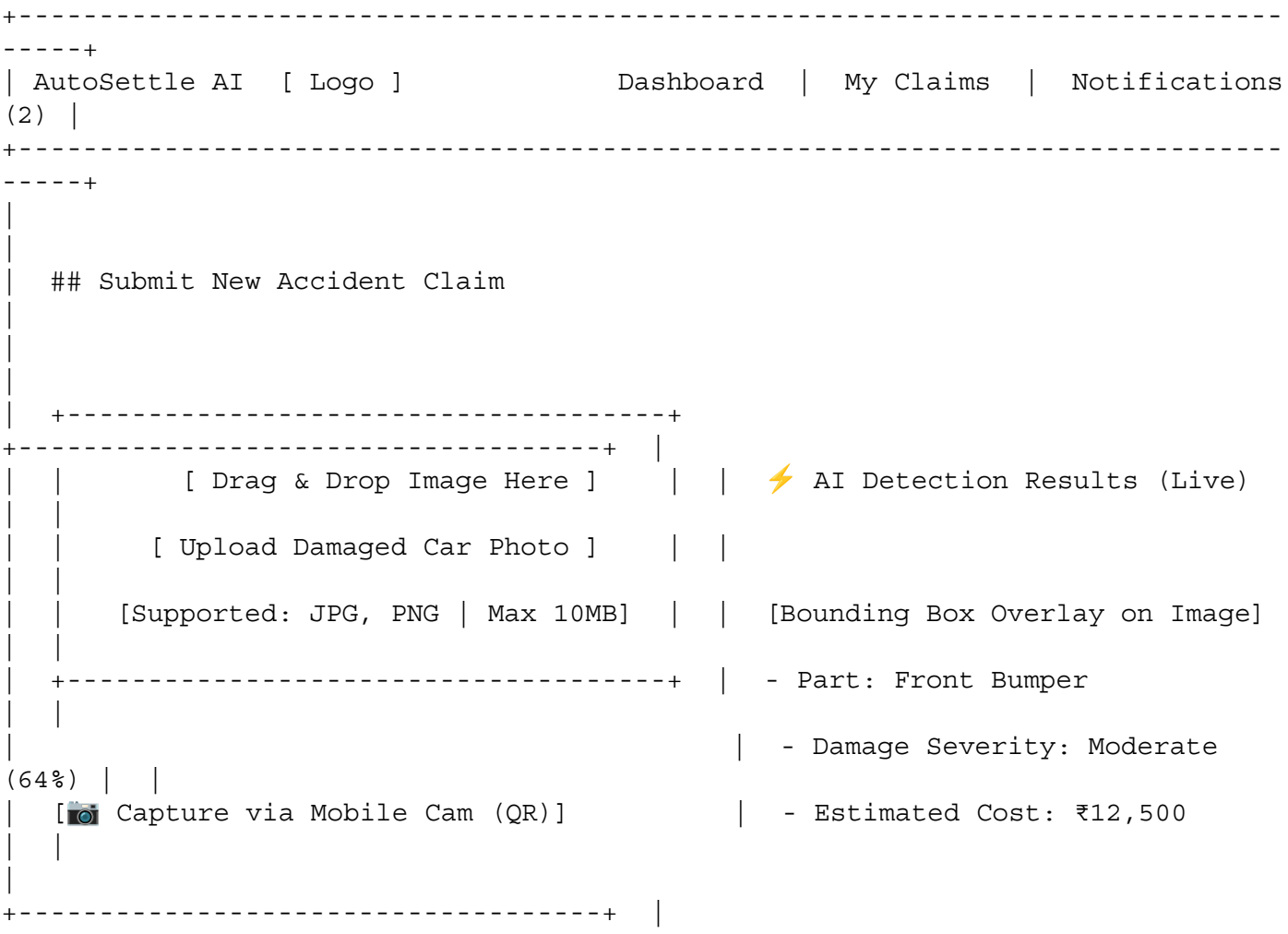
Response: AutoSettle AI: Figma Layout Wireframe

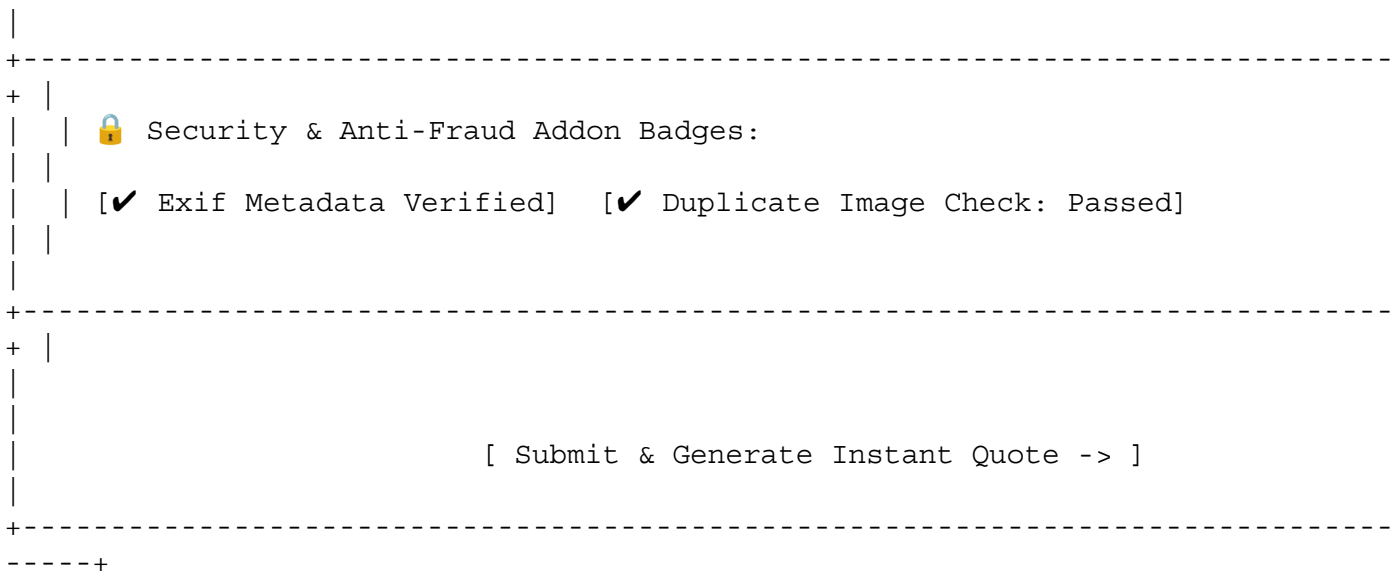
Here is a blueprint for your Figma layout. Since you are building a skeleton structure with high-impact addons, organizing your screens into a clear user flow will let your designers whip up mockups in minutes or help you code the UI instantly.

The app is divided into two primary dashboards: **User Portal** (for the driver/victim) and **Adjuster Desk** (for the insurance employee).

Screen 1: User Portal – Claim Submission & Live AI Analysis

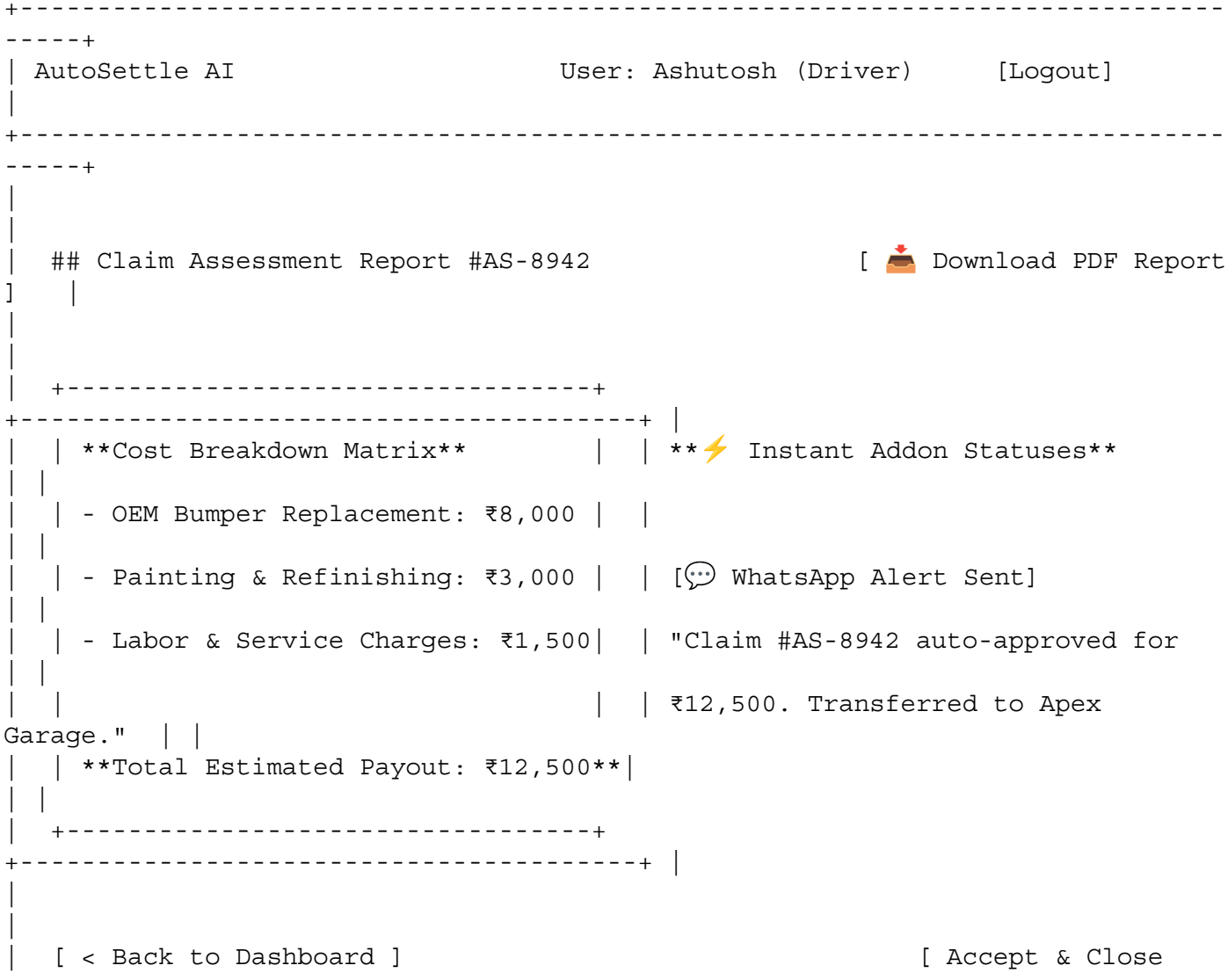
- **Goal:** Make photo upload and AI instant damage detection look seamless and high-tech.





Screen 2: User Portal – Interactive Estimate & Addon Notification

- **Goal:** Show the financial breakdown, PDF download, and WhatsApp alert addon.



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Claim ] |
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Screen 3: Insurance Adjuster Desk (Multi-Role Switcher View)

- **Goal:** Demonstrate enterprise value and how back-office employees review AI-flagged fraud or approve payouts in 1 click.

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| AutoSettle AI - ADMIN DESK          [ Role Switcher: User / Adjuster Desk ]
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|
|
|  ## Incoming Claims Queue (3 Pending Review)
|
|
|
|  | Claim ID | Vehicle | Damage Severity | AI Payout | Fraud Risk | Actions
|
|
|----------|---------|-----------------|-----------|------------|------------|
|  | #AS-8942 | Swift   | Moderate (64%)  | ₹12,500   | Low (0.02) | [Review]
|
|  | #AS-8943 | Creta   | Severe (89%)    | ₹45,000   | ⚠️ HIGH    | [Flag
Scam]|
|  | #AS-8944 | Baleno  | Minor (18%)     | ₹3,200    | Low (0.01) | [Review]
|
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|  | **Selected Claim Details (#AS-8943 - High Fraud Alert)**
| |
|  | - Reason flagged: Image hash matches a claim filed 4 months ago in Mumbai.
| |
|  | - Recommendation: Manual physical inspection required.
| |
|  |                                      [ Overrule AI ]    [ Dispatch
Inspector ] |
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